

Sidharth Rao

Naperville, IL | (630) 947-5839 | sidharthprao@gmail.com

Education

Iowa State University | Ames, Iowa

May 2026

Computer Engineering | GPA: 3.5/4.0 | Dean's List: F22, F23, S23

Relevant Coursework: Database Management Systems, Algorithms, Software Development, Linux OS Essentials, Embedded Systems, Network Protocols & Security

Experience

ARA Wireless Living Lab

May - Aug 2025

Undergraduate Research Assistant

- Assisted in designing and building User Equipment (UE) units for the ARA Wireless Living Lab to improve wireless connectivity and affordability in rural and agricultural areas.
- Automated setup of UE Linux computers by writing Ansible YAML playbooks for package installation and network configuration to expedite setup time of additional computers by up to 70% and standardize contents of several machines.
- Configured Quectel antennae using Minicom to communicate with the Iowa State ARA internal network.

Projects

MicroCART Quadcopter

Sep 2025 - Present

- Designed and developed an autonomous quadcopter system integrating a Raspberry Pi and camera module with a Crazyflie UAV platform for real-time computer vision-based navigation
- Implemented ArUco marker detection using OpenCV to enable precise autonomous landing, converting visual position data into quadcopter setpoints via a custom PID controller
- Engineered a custom quadcopter frame upgrade to improve structural robustness and flight performance, supporting additional onboard computing hardware

MIPS Processor

Sep – Nov 2024

- Designed and implemented a single-cycle MIPS processor in VHDL, integrating datapaths, control units, and ALU.
- Enhanced processor performance by developing hardware and software-scheduled pipelines to optimize instruction execution time by over double.
- Validated functionality through extensive simulation and testing using Questa Sim.

TuneFlow

Aug – Nov 2024

- Developed the frontend of an Android social media app using Java, providing song recommendations via Spotify based on friends' suggestions. Utilizing Java, using the Volley library for API requests, and incorporating testing using Postman, JUnit, and Mockito.
- Engineered a seamless integration with the Spotify Web API to gather and analyze user data, resulting in personalized song recommendations that improved engagement metrics by enhancing daily active users on the app by 40%.
- Designed and coded multiple unique user interfaces for the app, to increase potential user retention.

Skills

Java, C, Python, SQL, Jupyter, Git, LaTeX, XML, Ansible, Microsoft Office, G-Suite